

Physics 172

Chapter 8
Conservation Laws

Griffiths, D. *Introduction to Electrodynamics*.

Chapter 8: Conservation Laws

8.1 Charge and Energy

8.1.1 The Continuity Equation

The net charge in some volume bounded by S :

$$Q(t) = \int_V \rho(\mathbf{r}, t) d\tau$$

The net current through this volume :

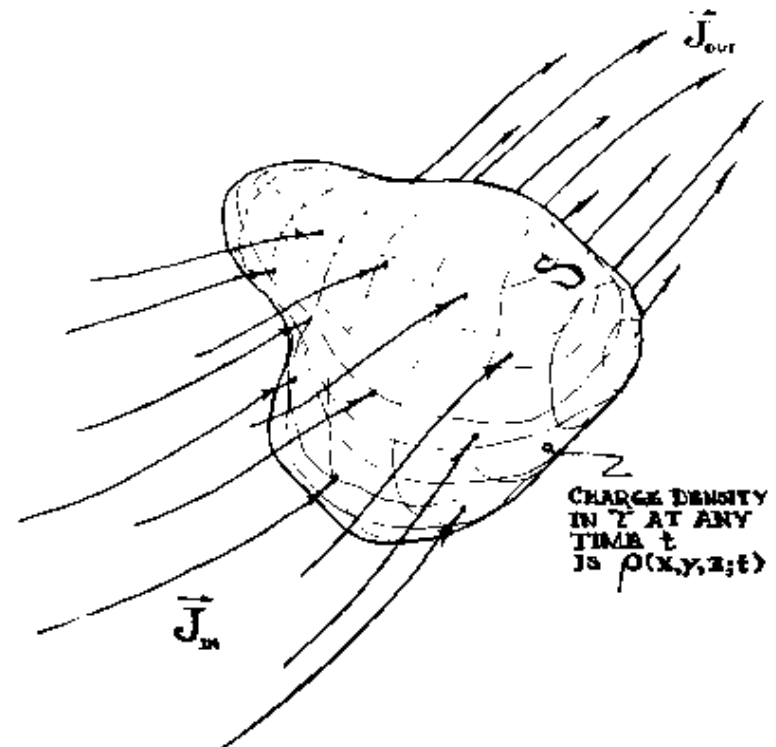
$$\frac{dQ}{dt} = - \int_S \mathbf{J} \cdot d\mathbf{a}$$

$$\frac{dQ}{dt} = \int_V \frac{\partial \rho}{\partial t} d\tau = - \int_V (\nabla \cdot \mathbf{J}) d\tau$$

(By Gauss' theorem)

The continuity equation :

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}$$



(8.4)

8.1.2 Poynting's Theorem

Energy stored in the fields :

$$W_e = \frac{\epsilon_0}{2} \int E^2 d\tau \qquad W_m = \frac{1}{2\mu_0} \int B^2 d\tau.$$

$$U_{em} = \frac{1}{2} \int \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau.$$

When charges move a bit :

$$dW_i = \mathbf{F} \cdot d\mathbf{l} = q_i (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = q_i (\mathbf{E} \cdot \mathbf{v}) dt$$

(with $q = \rho d\tau$ and $\rho \mathbf{v} = \mathbf{J}$,)

Power :

$$\frac{dW}{dt} = \int_V (\mathbf{E} \cdot \mathbf{J}) d\tau \qquad (8.6)$$

8.1.2 Poynting's Theorem

Power :

$$\frac{dW}{dt} = \int_V (\mathbf{E} \cdot \mathbf{J}) d\tau$$

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \quad \text{From } \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$= \frac{1}{\mu_0} [\mathbf{B} \cdot (\nabla \times \mathbf{E}) - \nabla \cdot (\mathbf{E} \times \mathbf{B})] - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}$$

$$\text{From } \nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}).$$

$$\mathbf{E} \cdot \mathbf{J} = \left(-\frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) \quad \text{From } \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$= -\left(\frac{1}{2\mu_0} \frac{\partial |\mathbf{B}|^2}{\partial t} + \frac{\epsilon_0}{2} \frac{\partial |\mathbf{E}|^2}{\partial t} \right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) \quad \text{From } \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (B^2)$$

8.1.2 Poynting's Theorem

Power :

$$\begin{aligned}\frac{dW}{dt} &= -\int_V \frac{\partial}{\partial t} \left(\frac{|\mathbf{B}|^2}{2\mu_0} + \frac{\varepsilon_0 |\mathbf{E}|^2}{2} \right) d\tau - \frac{1}{\mu_0} \int_V \nabla \cdot (\mathbf{E} \times \mathbf{B}) d\tau \\ &= -\frac{dU_{em}}{dt} - \frac{1}{\mu_0} \int_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a} \quad \text{With} \quad u_{em} = \frac{|\mathbf{B}|^2}{2\mu_0} + \frac{\varepsilon_0 |\mathbf{E}|^2}{2}\end{aligned}$$

- Define the Poynting vector :

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) \quad (8.10)$$

$$\frac{dW}{dt} = -\frac{dU_{em}}{dt} - \int_S \mathbf{S} \cdot d\mathbf{a} \quad \text{Poynting's theorem}$$

8.1.2 Poynting's Theorem

Poynting's theorem :

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B})$$

$$\frac{dW}{dt} = -\frac{dU_{em}}{dt} - \int_S \mathbf{S} \cdot d\mathbf{a}$$

- Total energy :
 - *Define the total energy density : $u = u_{mech} + u_{em}$*

$$\frac{dW}{dt} + \frac{dU_{em}}{dt} = -\int_V (\nabla \cdot \mathbf{S}) d\tau$$

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}$$

*Continuity equation
for Energy*

(8.12)

8.1.2 Poynting's Theorem

Example 8.1 :

Joule heating in a current carrying conductor.

- *Electric field:*

$$E = \frac{V}{L}$$

V = *potential difference*

- *Magnetic field :*

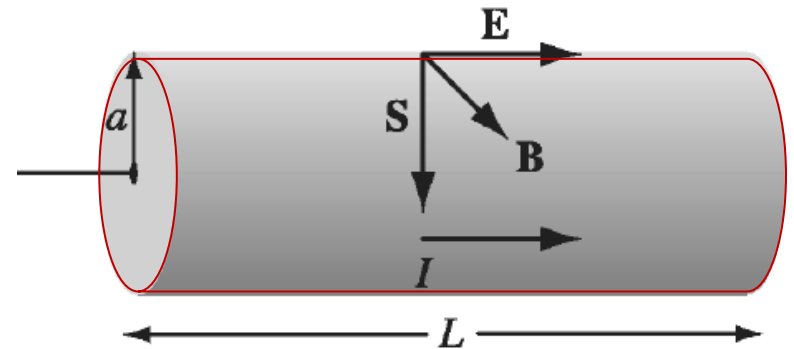
$$B = \frac{\mu_0 I}{2\pi a}$$

- *Poynting vector :*

$$S = -\frac{1}{\mu_0} \frac{V}{L} \frac{\mu_0 I}{2\pi a} = \frac{VI}{2\pi aL}$$

- *Power :*

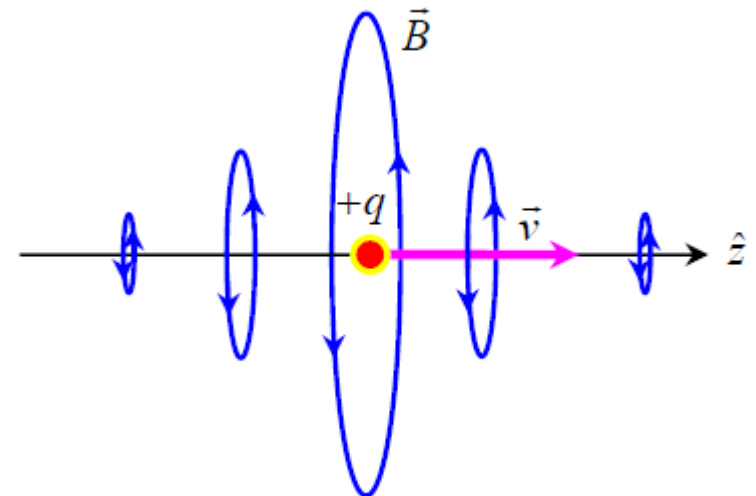
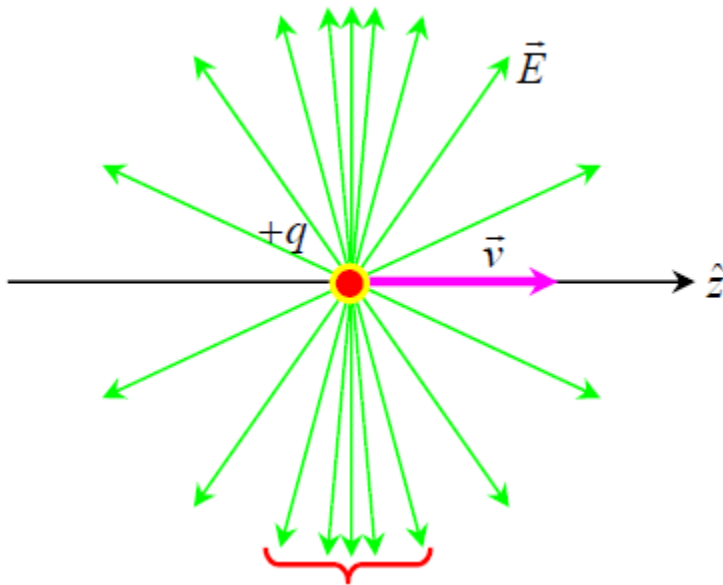
$$\frac{dU_{em}}{dt} = -\int_S \mathbf{S} \cdot d\mathbf{a} = S(2\pi aL) = VI$$



8.2 Momentum

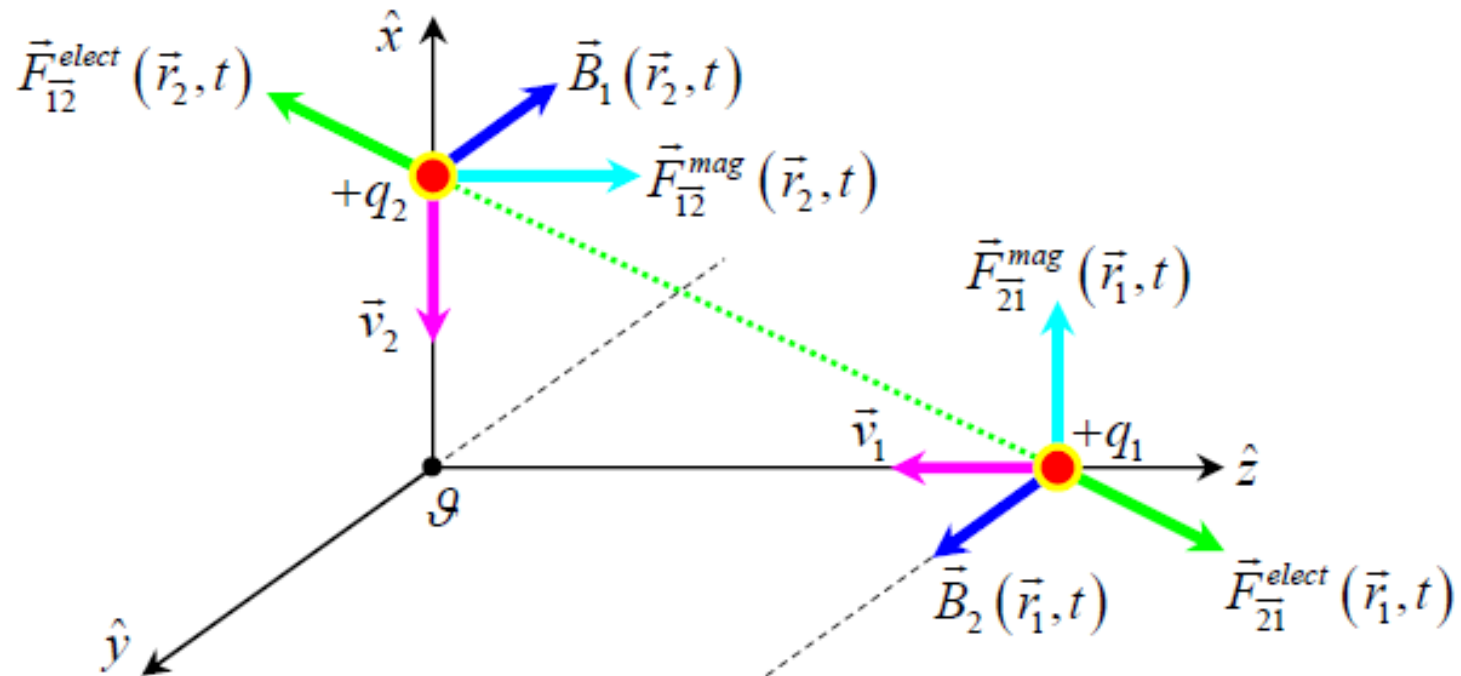
8.2.1 Newton's Third Law

Fields due to a moving point charge :



8.2.1 Newton's Third Law

Interaction between two moving point charges :



Is Newton's law of action-reaction violated ?

How about the conservation of momentum ?

8.2.2 Maxwell's Stress Tensor

Total electromagnetic force on a charge volume :

$$\mathbf{F} = \int_V (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \rho d\tau = \int_V (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) d\tau$$

- The force per unit volume : $\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}$

$$\mathbf{f} = \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B}$$

$$\begin{aligned} & \text{(from } \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\ & \text{and } \rho = \epsilon_0 \nabla \cdot \mathbf{E} \text{)} \end{aligned}$$

$$= \epsilon_0 [(\nabla \cdot \mathbf{E}) \mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] - \frac{1}{\mu_0} [\mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B})$$

$$\begin{aligned} & \text{(from } \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) = \frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} + \mathbf{E} \times \frac{\partial \mathbf{B}}{\partial t} \quad \text{and} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \text{)} \end{aligned}$$

8.2.2 Maxwell's Stress Tensor

Total electromagnetic force on a charge volume :

$$\mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] - \frac{1}{\mu_0} [\mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B})$$

- Adding the term : $(\nabla \cdot \mathbf{B})\mathbf{B}$

$$\mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} - \mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B})$$

$$(\text{from } \nabla(E^2) = 2(\mathbf{E} \cdot \nabla)\mathbf{E} + 2\mathbf{E} \times (\nabla \times \mathbf{E}) \text{ and } \mathbf{E} \times (\nabla \times \mathbf{E}) = \frac{1}{2}\nabla(E^2) - (\mathbf{E} \cdot \nabla)\mathbf{E},)$$

$$\mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}]$$

$$- \frac{1}{2}\nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}).$$

(8.16)

8.2.2 Maxwell's Stress Tensor

The force per unit volume :

$$\mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] \\ - \frac{1}{2} \nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}).$$

- Introduce the **Maxwell Stress Tensor** :

$$T_{ij} \equiv \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right) \quad (8.17)$$

$$\vec{\mathbf{T}} = \left(\epsilon_0 \mathbf{E}\mathbf{E} + \frac{1}{\mu_0} \mathbf{B}\mathbf{B} \right) - \frac{1}{2} \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \vec{\mathbf{I}}$$

8.2.2 Maxwell's Stress Tensor

*Dyad product and tensors :

- For two vectors : $\mathbf{a} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} + a_z \hat{\mathbf{k}}$ $\mathbf{b} = b_x \hat{\mathbf{i}} + b_y \hat{\mathbf{j}} + b_z \hat{\mathbf{k}}$

$$\begin{aligned} \mathbf{ab} \equiv & a_x b_x \hat{\mathbf{i}}\hat{\mathbf{i}} + a_y b_x \hat{\mathbf{j}}\hat{\mathbf{i}} + a_z b_x \hat{\mathbf{k}}\hat{\mathbf{i}} + \\ & a_x b_y \hat{\mathbf{i}}\hat{\mathbf{j}} + a_y b_y \hat{\mathbf{j}}\hat{\mathbf{j}} + a_z b_y \hat{\mathbf{k}}\hat{\mathbf{j}} + \\ & a_x b_z \hat{\mathbf{i}}\hat{\mathbf{k}} + a_y b_z \hat{\mathbf{j}}\hat{\mathbf{k}} + a_z b_z \hat{\mathbf{k}}\hat{\mathbf{k}} \end{aligned} \quad (\text{Also: } \mathbf{a} * \mathbf{b})$$

- In matrix form :

$$\mathbf{ab} = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix} \quad \text{or} \quad \mathbf{ab} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} \begin{pmatrix} b_x & b_y & b_z \end{pmatrix}$$

8.2.2 Maxwell's Stress Tensor

*Dyad product and tensors :

- The tensor :*

$$\vec{\mathbf{T}} = \begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{pmatrix} = \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix} \quad \vec{\mathbf{I}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(Unit Tensor)

- Pre-dot product of vector and tensor :*

$$\mathbf{a} \cdot \vec{\mathbf{T}} = \begin{pmatrix} a_x & a_y & a_z \end{pmatrix} \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix}$$

$$\left(\mathbf{a} \cdot \vec{\mathbf{T}} \right)_j = \sum_{i=x,y,z} a_i T_{ij}$$

8.2.2 Maxwell's Stress Tensor

*Dyad product and tensors :

- *Post-dot product of vector and tensor :*

$$\vec{\mathbf{T}} \cdot \mathbf{a} = \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix} \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} \quad \left(\vec{\mathbf{T}} \cdot \mathbf{a} \right)_i = \sum_{j=x,y,z} T_{ij} a_j$$

- *For the vector operator :*

$$\nabla \cdot \vec{\mathbf{T}} = \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix}$$

$$\left(\nabla \cdot \vec{\mathbf{T}} \right)_j = \frac{\partial T_{xj}}{\partial x} + \frac{\partial T_{yj}}{\partial y} + \frac{\partial T_{zj}}{\partial z}$$

8.2.2 Maxwell's Stress Tensor

*Dyad product and tensors :

- *Properties :* For vectors **a**, **b**, **c**, **d**, **w**

$$\mathbf{ab} \neq \mathbf{ba}$$

Dyads are **not** commutative

$$\mathbf{a}(\mathbf{b} + \mathbf{c}) = \mathbf{ab} + \mathbf{ac}$$

Distributive law

$$(\mathbf{a} + \mathbf{b})(\mathbf{c} + \mathbf{d}) = \mathbf{ac} + \mathbf{ad} + \mathbf{bc} + \mathbf{db}$$

Distributive law

$$\mathbf{w} \cdot (\mathbf{ab} + \mathbf{cd}) = (\mathbf{w} \cdot \mathbf{a})\mathbf{b} + (\mathbf{w} \cdot \mathbf{c})\mathbf{d}$$

Pre-dot product

$$(\mathbf{ab} + \mathbf{cd}) \cdot \mathbf{w} = \mathbf{a}(\mathbf{b} \cdot \mathbf{w}) + \mathbf{c}(\mathbf{d} \cdot \mathbf{w})$$

Post-dot product

$$\mathbf{w} \times (\mathbf{ab} + \mathbf{cd}) = (\mathbf{w} \times \mathbf{a})\mathbf{b} + (\mathbf{w} \times \mathbf{c})\mathbf{d}$$

Pre-cross product

$$(\mathbf{ab} + \mathbf{cd}) \times \mathbf{w} = \mathbf{a}(\mathbf{b} \times \mathbf{w}) + \mathbf{c}(\mathbf{d} \times \mathbf{w})$$

Post-cross product

8.2.2 Maxwell's Stress Tensor

Back to the stress tensor :

$$\vec{\mathbf{T}} = \left(\epsilon_0 \mathbf{E}\mathbf{E} + \frac{1}{\mu_0} \mathbf{B}\mathbf{B} \right) - \frac{1}{2} \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \vec{\mathbf{I}}$$

Or :

$$T_{ij} \equiv \epsilon_o \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_o} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right)$$

$$T_{xx} = \frac{1}{2} \epsilon_o (E_x^2 - E_y^2 - E_z^2) + \frac{1}{2\mu_o} (B_x^2 - B_y^2 - B_z^2) \quad T_{xy} = T_{yx} = \epsilon_o (E_x E_y) + \frac{1}{\mu_o} (B_x B_y)$$

$$T_{yy} = \frac{1}{2} \epsilon_o (E_y^2 - E_z^2 - E_x^2) + \frac{1}{2\mu_o} (B_y^2 - B_z^2 - B_x^2) \quad T_{xz} = T_{zx} = \epsilon_o (E_x E_z) + \frac{1}{\mu_o} (B_x B_z)$$

$$T_{zz} = \frac{1}{2} \epsilon_o (E_z^2 - E_x^2 - E_y^2) + \frac{1}{2\mu_o} (B_z^2 - B_x^2 - B_y^2) \quad T_{yz} = T_{zy} = \epsilon_o (E_y E_z) + \frac{1}{\mu_o} (B_y B_z)$$

8.2.2 Maxwell's Stress Tensor

Back to the stress tensor :

$$\vec{\mathbf{T}} = \left(\epsilon_0 \mathbf{E}\mathbf{E} + \frac{1}{\mu_0} \mathbf{B}\mathbf{B} \right) - \frac{1}{2} \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \vec{\mathbf{I}}$$

$$\nabla \cdot \vec{\mathbf{T}} = \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] - \frac{1}{2} \nabla \left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right)$$

- *The force density :*

$$\begin{aligned} \mathbf{f} = & \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] \\ & - \frac{1}{2} \nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \end{aligned}$$

or :

$$\mathbf{f} = \nabla \cdot \vec{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}$$

(8.19)

8.2.2 Maxwell's Stress Tensor

Back to the stress tensor :

$$T_{ij} \equiv \epsilon_o \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_o} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right)$$

- Then :

$$\begin{aligned} (\nabla \cdot \hat{\mathbf{T}})_j &= \epsilon_0 \left[(\nabla \cdot \mathbf{E}) E_j + (\mathbf{E} \cdot \nabla) E_j - \frac{1}{2} \nabla_j E^2 \right] \\ &+ \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) B_j + (\mathbf{B} \cdot \nabla) B_j - \frac{1}{2} \nabla_j B^2 \right] \end{aligned}$$

$$\mathbf{f} = \nabla \cdot \hat{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}$$

8.2.2 Maxwell's Stress Tensor

The total force :

$$\mathbf{f} = \nabla \cdot \vec{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}$$

$$\mathbf{F} = \int_V (\nabla \cdot \vec{\mathbf{T}}) d\tau - \epsilon_0 \mu_0 \int_V \frac{\partial \mathbf{S}}{\partial t} d\tau$$

Or :

$$\mathbf{F} = \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_V \mathbf{S} d\tau \quad (8.20)$$

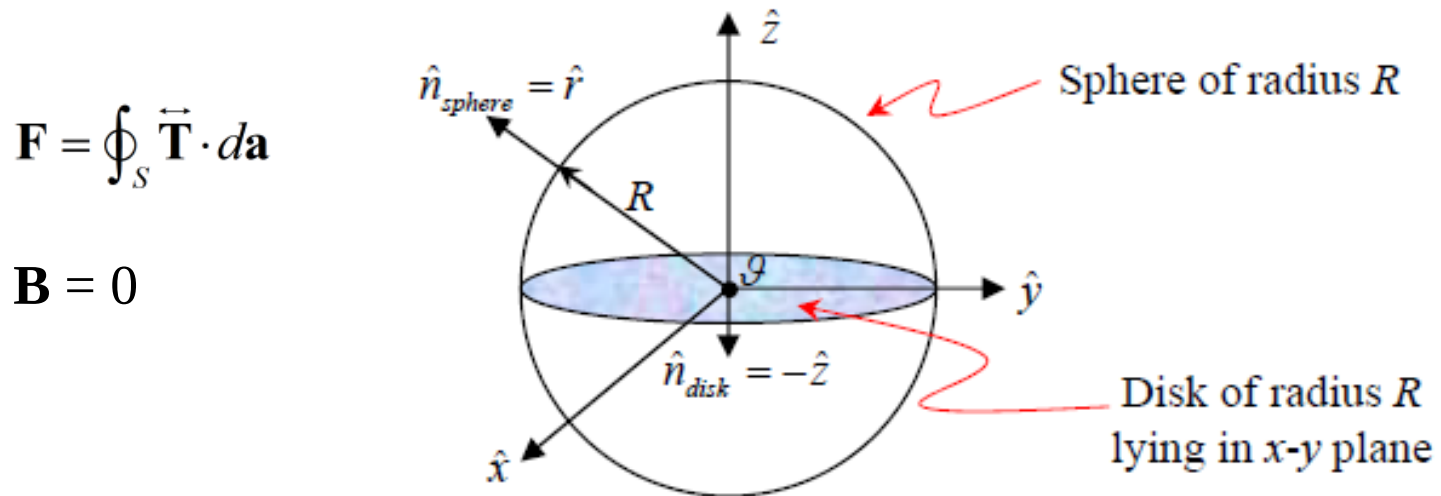
- For static fields :

$$\mathbf{F} = \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a}$$

8.2.2 Maxwell's Stress Tensor

Example 8.2 :

Determine the net force on the "northern" hemisphere of a uniformly charged solid sphere of radius R and charge Q (the same as Prob. 2.47, only this time we'll use the Maxwell stress tensor and Eq. 8.21).



$$\mathbf{F} = \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a}$$

$$\mathbf{B} = 0$$

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{\mathbf{r}} = \frac{Q}{4\pi\epsilon_0 R^2} [(\sin\theta \cos\phi)\hat{\mathbf{x}} + (\sin\theta \sin\phi)\hat{\mathbf{y}} + (\cos\theta)\hat{\mathbf{z}}]$$

$$d\mathbf{a} = R^2 \sin\theta d\theta d\phi \hat{\mathbf{r}} = R^2 d\theta d\phi [(\sin^2\theta \cos\phi)\hat{\mathbf{x}} + (\sin^2\theta \sin\phi)\hat{\mathbf{y}} + (\sin\theta \cos\theta)\hat{\mathbf{z}}]$$

8.2.2 Maxwell's Stress Tensor

Example 8.2 :

a. For upper "bowl" :

$$d\mathbf{a} = R^2 d\theta d\phi [(\sin^2 \theta \cos \phi) \hat{\mathbf{x}} + (\sin^2 \theta \sin \phi) \hat{\mathbf{y}} + (\sin \theta \cos \theta) \hat{\mathbf{z}}]$$

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} [(\sin \theta \cos \phi) \hat{\mathbf{x}} + (\sin \theta \sin \phi) \hat{\mathbf{y}} + (\cos \theta) \hat{\mathbf{z}}]$$

- Net force is in z-direction:

$$(\hat{\mathbf{T}} \cdot d\mathbf{a})_z = T_{zx} da_x + T_{zy} da_y + T_{zz} da_z$$

$$T_{zx} = \epsilon_0 E_z E_x = \epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 \sin \theta \cos \theta \cos \phi,$$

$$T_{zy} = \epsilon_0 E_z E_y = \epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 \sin \theta \cos \theta \sin \phi,$$

$$T_{zz} = \frac{\epsilon_0}{2} (E_z^2 - E_x^2 - E_y^2) = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^2} \right)^2 (\cos^2 \theta - \sin^2 \theta).$$

8.2.2 Maxwell's Stress Tensor

Example 8.2 :

a. For upper "bowl" :

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = T_{zx} da_x + T_{zy} da_y + T_{zz} da_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 \sin\theta \cos\theta d\theta d\phi.$$

$$F_{\text{bowl}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 2\pi \int_0^{\pi/2} \sin\theta \cos\theta d\theta = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2}$$

b. For equatorial "disc" :

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \mathbf{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} r (\cos\phi \hat{\mathbf{x}} + \sin\phi \hat{\mathbf{y}}), \quad d\mathbf{a} = -r dr d\phi \hat{\mathbf{z}}$$

$$T_{zz} = \frac{\epsilon_0}{2} (E_z^2 - E_x^2 - E_y^2) = -\frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 r^2.$$

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 r^3 dr d\phi$$

8.2.2 Maxwell's Stress Tensor

Example 8.2 :

a. For upper “bowl” :

$$F_{\text{bowl}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 2\pi \int_0^{\pi/2} \sin\theta \cos\theta d\theta = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2}$$

b. For equatorial “disc” :

$$(\vec{\mathbf{T}} \cdot d\mathbf{a})_z = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 r^3 dr d\phi$$

$$F_{\text{disk}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 2\pi \int_0^R r^3 dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{16R^2}$$

8.2.2 Maxwell's Stress Tensor

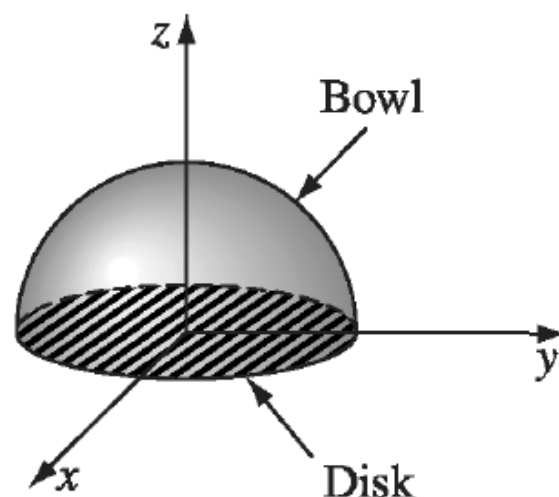
Example 8.2 :

$$F_{\text{bowl}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R} \right)^2 2\pi \int_0^{\pi/2} \sin\theta \cos\theta d\theta = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{8R^2}$$

$$F_{\text{disk}} = \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 R^3} \right)^2 2\pi \int_0^R r^3 dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{16R^2}$$

• Total force :

$$F = \frac{1}{4\pi\epsilon_0} \frac{3Q^2}{16R^2}$$



8.2.3 Conservation of Momentum

From Newton's second law :

$$\mathbf{F} = \frac{d\mathbf{p}_{\text{mech}}}{dt} = -\epsilon_0\mu_0 \frac{d}{dt} \int_V \mathbf{S} d\tau + \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a} \quad (8.27)$$

- Identify momentum stored in electromagnetic fields :

$$\mathbf{p}_{\text{em}} = \mu_0\epsilon_0 \int_V \mathbf{S} d\tau, \quad = \text{Momentum stored in the fields.}$$

$$\oint_S \vec{\mathbf{T}} \cdot d\mathbf{a}, \quad = \text{Momentum flowing through the surface per unit time.}$$

$$\frac{d}{dt}(\mathbf{p}_{\text{mech}} + \mathbf{p}_{\text{em}}) = \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a}$$

8.2.3 Conservation of Momentum

Momentum density :

$$\frac{d}{dt}(\mathbf{p}_{mech} + \mathbf{p}_{em}) = \oint_S \vec{\mathbf{T}} \cdot d\mathbf{a}$$

- Define $\mathcal{P}$:

$$\mathbf{p} = \int_V \mathcal{P} d\tau \quad \wp_{em} = \mu_0 \epsilon_0 \mathbf{S} = \text{Momentum density.}$$

$$-\vec{\mathbf{T}} = \text{Momentum flux density.}$$

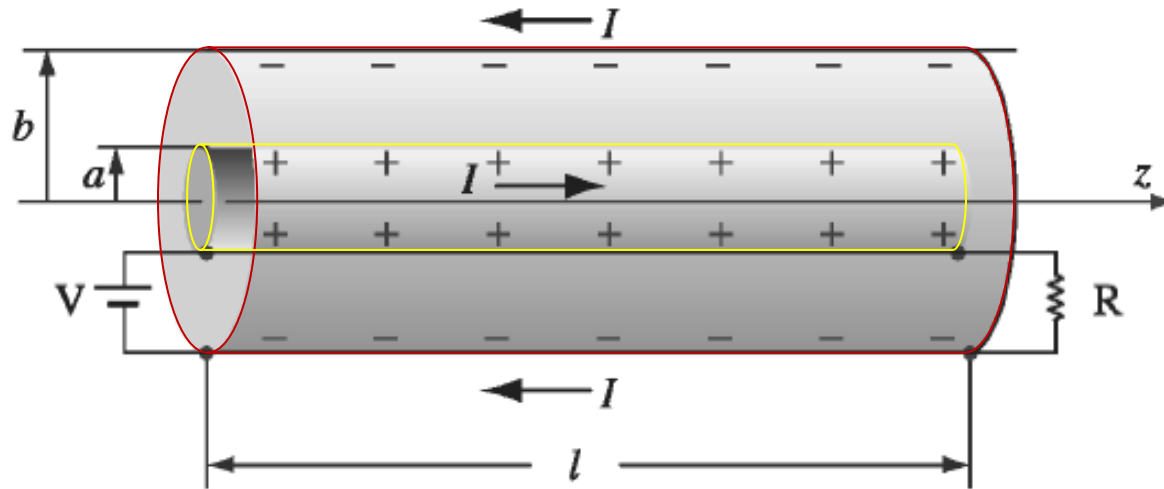
- Conservation principle:

$$\frac{\partial}{\partial t}(\wp_{mech} + \wp_{em}) = \nabla \cdot \vec{\mathbf{T}} \quad (8.30)$$

8.2.3 Conservation of Momentum

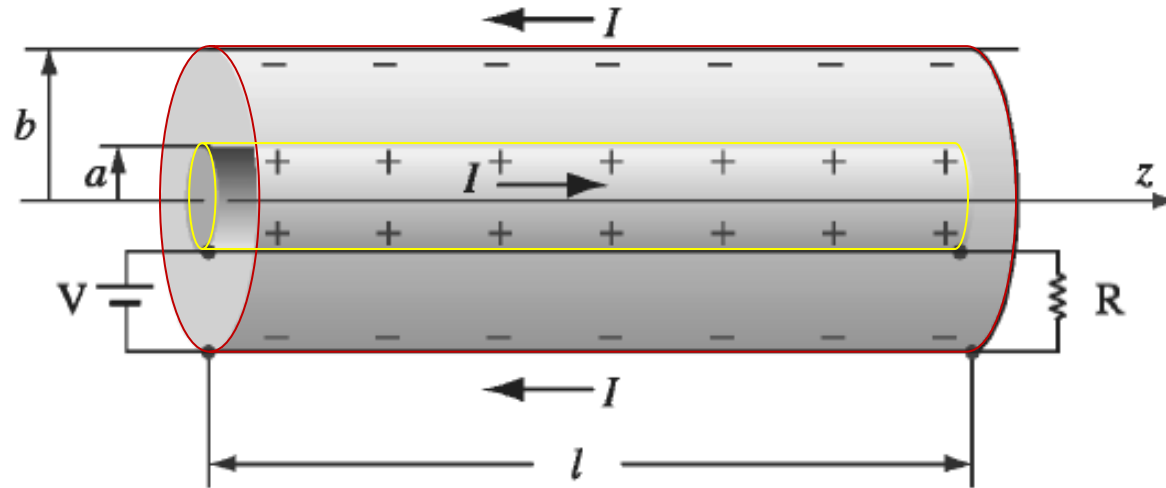
Example 8.3 :

A long coaxial cable, of length l , consists of an inner conductor (radius a) and an outer conductor (radius b). It is connected to a battery at one end and a resistor at the other. The inner conductor carries a uniform charge per unit length λ , and a steady current I to the right; the outer conductor has the opposite charge and current. What is the electromagnetic momentum stored in the fields?



8.2.3 Conservation of Momentum

Example 8.3 :



Solution:

- The fields:*

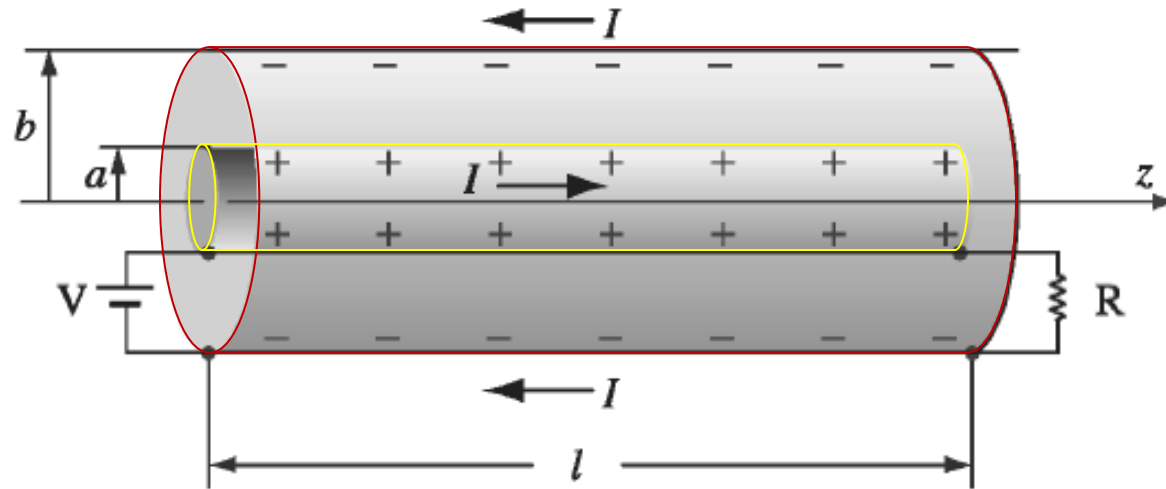
$$\mathbf{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{s} \hat{\mathbf{s}}, \quad \mathbf{B} = \frac{\mu_0}{2\pi} \frac{I}{s} \hat{\boldsymbol{\phi}}.$$

- The Poynting vector:*

$$\mathbf{S} = \frac{\lambda I}{4\pi^2\epsilon_0 s^2} \hat{\mathbf{z}}.$$

8.2.3 Conservation of Momentum

Example 8.3 :



Solution:

- The power transported:*

$$P = \int \mathbf{S} \cdot d\mathbf{a} = \frac{\lambda I}{4\pi^2 \epsilon_0} \int_a^b \frac{1}{s^2} 2\pi s ds = \frac{\lambda I}{2\pi \epsilon_0} \ln(b/a) = IV,$$

- The momentum:*

$$\mathbf{p} = \mu_0 \epsilon_0 \int \mathbf{S} d\tau = \frac{\mu_0 \lambda I}{4\pi^2} \hat{\mathbf{z}} \int_a^b \frac{1}{s^2} l 2\pi s ds = \frac{\mu_0 \lambda I l}{2\pi} \ln(b/a) \hat{\mathbf{z}} = \frac{IVl}{c^2} \hat{\mathbf{z}}.$$

8.2.4 Angular Momentum

EM-fields have energy and momentum :

- *Energy density:*

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

- *Momentum density:*

$$\mathbf{g} = \epsilon_0 (\mathbf{E} \times \mathbf{B}),$$

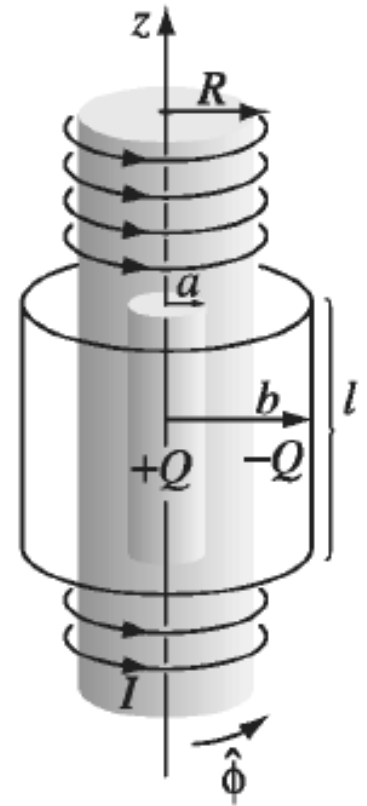
- *Angular momentum density :*

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \epsilon_0 [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})] \quad (8.33)$$

8.2.4 Angular Momentum

Example 8.3 :

Imagine a very long solenoid with radius R , n turns per unit length, and current I . Coaxial with the solenoid are two long cylindrical (nonconducting) shells of length l – one, *inside* the solenoid at radius a , carries a charge $+Q$, uniformly distributed over its surface; the other, *outside* the solenoid at radius b , carries charge $-Q$ (l is supposed to be much greater than b). When the current in the solenoid is gradually reduced, the cylinders begin to rotate. *Question:* Where does the angular momentum come from?



8.2.4 Angular Momentum

Example 8.3 :

Solution:

- Initial fields:*

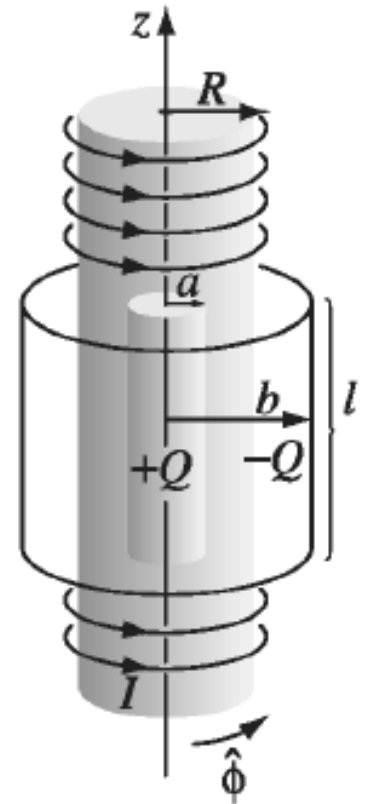
$$\mathbf{E} = \frac{Q}{2\pi\epsilon_0 l s} \hat{\mathbf{s}} \quad (a < s < b), \quad \mathbf{B} = \mu_0 n I \hat{\mathbf{z}} \quad (s < R)$$

- Momentum densities :*

$$\mathbf{g} = -\frac{\mu_0 n I Q}{2\pi l s} \hat{\boldsymbol{\phi}} \quad (\mathbf{r} \times \mathbf{g})_z = -\frac{\mu_0 n I Q}{2\pi l}$$

- The angular momentum:*

$$\mathbf{L} = -\frac{1}{2} \mu_0 n I Q (R^2 - a^2) \hat{\mathbf{z}}.$$



8.2.4 Angular Momentum

Example 8.3 :

Solution:

- When the current is turned off, induced electric field:

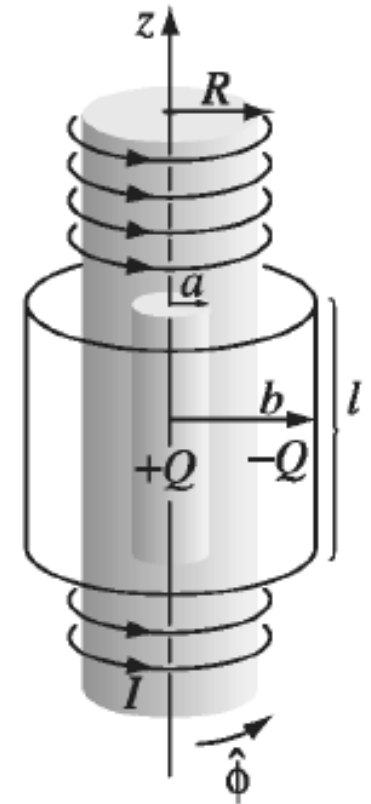
$$\mathbf{E} = \begin{cases} -\frac{1}{2}\mu_0 n \frac{dI}{dt} \frac{R^2}{s} \hat{\phi}, & (s > R), \\ -\frac{1}{2}\mu_0 n \frac{dI}{dt} s \hat{\phi}, & (s < R). \end{cases}$$

- Force on Q produces torque:

$$\mathbf{N}_b = \mathbf{r} \times (-Q\mathbf{E}) = \frac{1}{2}\mu_0 n Q R^2 \frac{dI}{dt} \hat{\mathbf{z}}, \quad \text{On outer cylinder.}$$

$$\mathbf{N}_a = -\frac{1}{2}\mu_0 n Q a^2 \frac{dI}{dt} \hat{\mathbf{z}}. \quad \text{On inner cylinder.}$$

$$\mathbf{L} = -\frac{1}{2}\mu_0 n I Q (R^2 - a^2) \hat{\mathbf{z}}.$$



8.2.4 Angular Momentum

Example 8.3 :

Solution:

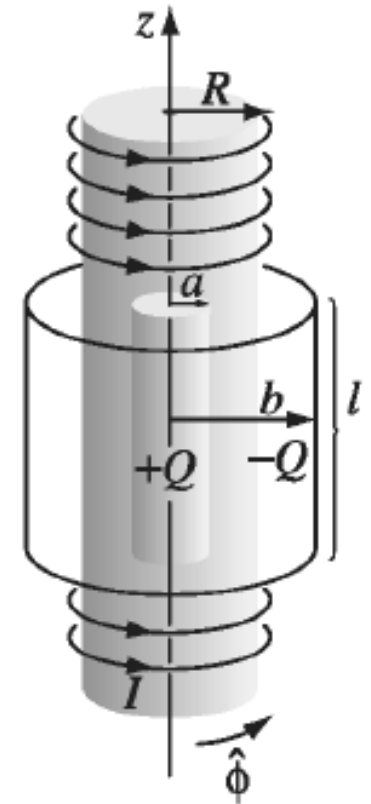
- Torque produces angular momentum:*

$$\mathbf{L}_b = \frac{1}{2} \mu_0 n Q R^2 \hat{\mathbf{z}} \int_I \frac{dI}{dt} dt = -\frac{1}{2} \mu_0 n I Q R^2 \hat{\mathbf{z}}. \quad \text{On outer cylinder.}$$

$$\mathbf{L}_a = \frac{1}{2} \mu_0 n I Q a^2 \hat{\mathbf{z}}. \quad \text{On inner cylinder.}$$

- Total angular momentum:*

$$\mathbf{L}_{\text{em}} = \mathbf{L}_a + \mathbf{L}_b.$$



$$\mathbf{L} = -\frac{1}{2} \mu_0 n I Q (R^2 - a^2) \hat{\mathbf{z}}.$$

Chapter 8

End